

Lutetium

Lu (Lu I). Strongest classified transitions below 2 μm (NIST): 41 levels, 66 lines.

Lu⁺ ionisation limit 43762.60 cm⁻¹ = 5.42587 eV (228.506 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element

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J
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e

r

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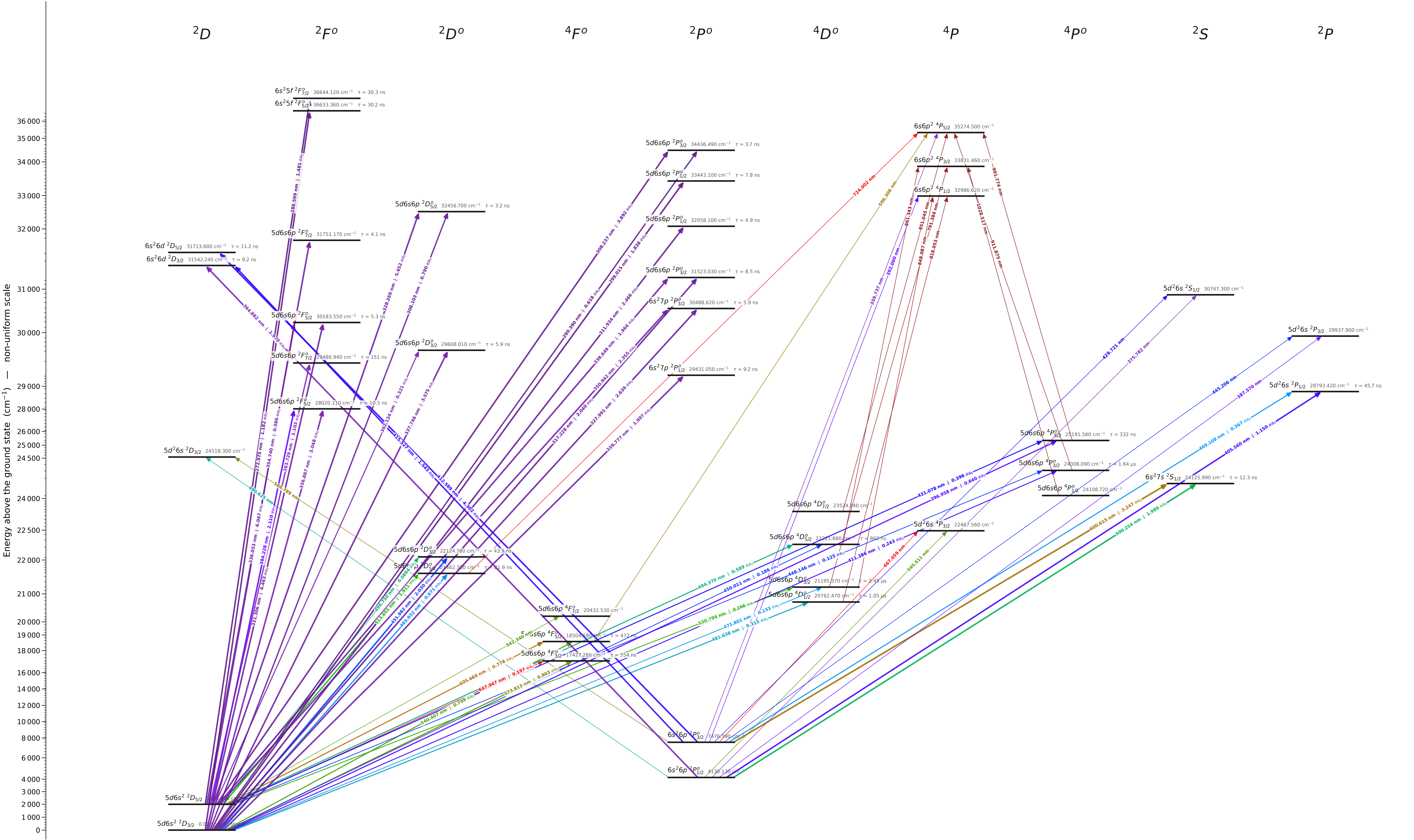
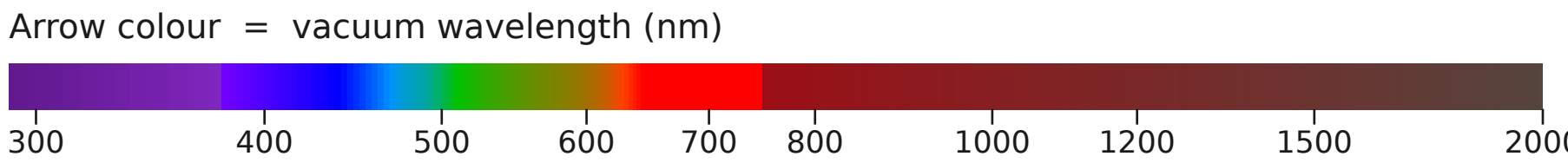
J
′

⟩

|

{\displaystyle |\langle J|er|J'\rangle |}

 in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	 ⟨ J e r J ′ ⟩ {\displaystyle \langle J er J'\rangle } (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5d6s ² ² D _{3/2} – 5d6s6p ² D _{3/2} ^o	451.9823	451.8556	663.2836	2.030	2.260e+07	3.625	99.4	451.9 nm line of collinear laser spectroscopy at ISOLDE
5d6s ² ² D _{3/2} – 5d6s6p ² D _{5/2} ^o	465.9316	465.8011	643.4260	0.971	3.150e+06	1.943	25.8	first step
5d6s ² ² D _{3/2} – 5d6s6p ⁴ F _{3/2} ^o	540.4068	540.2565	554.7533	0.749	1.200e+06	0.3372	56.6	first step
5d6s ² ² D _{3/2} – 5d6s6p ⁴ F _{3/2} ^o	573.8130	573.6539	522.4567	0.803	1.730e+06	0.2873	96	first step

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
5d6s ² ² D _{3/2} – 5d6s6p ² D _{3/2} ^o	451.982	175-176	435(7)

Shift = ν(first isotope) – ν(second isotope).

¹⁷⁵Lu hyperfine structure (I = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
5d6s ² ² D _{3/2}	194.33162(9)	1511.3987(7)	F=2: -502.06 F=3: -998.64 F=4: -653.14 F=5: +1398.09
5d6s ² ² D _{5/2}	146.776	1860.656	F=1: -654.46 F=2: -892.52 F=3: -1050.26 F=4: -888.44 F=5: -88.11 F=6: +1749.46
6s ² 6p ² P _{3/2} ^o	221(8)	2091(23)	F=2: -370.22 F=3: -1201.39 F=4: -915.62 F=5: +1681.95
5d6s6p ⁴ F _{3/2} ^o	-924.7(5)	1767(6)	F=2: +7188.33 F=3: +3152.09 F=4: -1051.57 F=5: -4412.93
5d6s6p ⁴ F _{5/2} ^o	987.2(4)	1117(6)	F=1: -10507.61 F=2: -8852.35 F=3: -6249.79 F=4: -2556.30 F=5: +2419.59 F=6: +8917.25
5d6s6p ⁴ F _{7/2} ^o	1016(4)	2531(22)	F=0: -14653.98 F=1: -13895.75 F=2: -12327.62 F=3: -9846.31 F=4: -6296.84 F=5: -1472.61 F=6: +4884.66 F=7: +13084.88
5d6s6p ⁴ D _{3/2} ^o	-2244(3)	-	F=3: +5049.00 F=4: -3927.00
5d6s6p ⁴ D _{5/2} ^o	1026(5)	615(14)	F=2: -6595.36 F=3: -3956.95 F=4: -29.06 F=5: +5539.73
5d6s6p ² D _{3/2} ^o	1398.2(15)	-689(30)	F=1: -16098.86 F=2: -13105.60 F=3: -8689.54 F=4: -2939.25 F=5: +4027.14 F=6: +12062.00
5d6s6p ² D _{5/2} ^o	-1424.7(7)	-501(6)	F=2: +9348.33 F=3: +5432.09 F=4: -123.57 F=5: -7604.93
5d6s6p ⁴ D _{5/2} ^o	1092(3)	855(30)	F=1: -11826.96 F=2: -9887.25 F=3: -6886.07 F=4: -2713.50 F=5: +2777.04 F=6: +9768.75
6s ² 7s ² S _{1/2}	1170(3)	-	F=3: -2632.50 F=4: +2047.50
5d ² 6s ² P _{3/2}	-1021(3)	-1190(15)	F=2: +6254.25 F=3: +4041.25 F=4: +297.25 F=5: -5657.75
6s6p ² ⁴ P _{5/2}	1933.660	-2509.260	F=1: -23097.92 F=2: -18513.67 F=3: -11906.14 F=4: -3597.96 F=5: +5980.73 F=6: +16292.21

Measured A, B constants; sources in the data file.